

Grade 11 Functions

Quadratic Function

A quadratic function is any polynomial function of order 2. (This means that the highest exponent of a variable in the function is 2 and that all exponents of variables are natural numbers)

Last year we studied quadratic functions of the form: $y = ax^2 + q$.

This year we will be studying quadratic functions of the form: $y = a(x - p)^2 + q$.

Use Geogebra to open the “Grade 11 Parabola Investigation”. You can do this by using ANY OF the 3 methods listed below.

Method 1: Open up a web browser and go to <http://tube.geogebra.org/student/m949871> (be careful if you use the mouse scroll wheel you will zoom in and out – if you want to move the web page up and down use the bar on the right of the screen).

Method 2: Copy the folder called “Investigation 1 – Quadratic Function” on the S drive found at [S:\Grade11\Maths and APM\Functions Investigation](#). Go into the folder and open up the web page called “Grade 11 Parabola Investigation”. You might need to use allow javascript to run. Try Google Chrome if it won’t run in Internet Explorer.

Method 3: Install Geogebra (you can find an installer at [S:\Grade11\Maths and APM\Functions Investigation\Geogebra Files](#) or you can download it from the internet). Copy the file called “Grade 11 Functions - 1 - Parabolas - Geogebra file” onto your flash drive and run it once you have installed Geogebra. The downstairs high school computer lab has Geogebra already installed.

Once you have Geogebra running you should see a Cartesian plane, a blue line running across the screen and 3 sliders, labelled a , p and q . You should also see the equation of the function written in blue. **Ensure that a , p and q are all set to 0.**

1. Look at the equation of the function, it should be $f(x) = 0(x)^2$. Why is the function a flat line?

2. Keep $a = 0$, adjust the value of the slider q . Describe how the graph changes.

3. Keep $a = 0$, adjust the value of the slider p . Does the graph look like it is changing?

4. Reset p and q to 0. Adjust the value of a so that $a > 0$.

a) Describe what happens to the graph.

b) What is the difference in the graph when a is close to 0 and when a is close to 5?

5. Repeat step 4, but this time adjust the value of a so that $a < 0$.

a) Describe what happens to the graph.

b) What is the difference in the graph when a is close to 0 and when a is close to -5 ?

6. Ensure that a is NOT 0. Keep $p = 0$. Adjust the value of q .

a) Describe how the function text changes (the text is in blue underneath the sliders).

b) Describe how the graph (the picture) changes as q changes. Be precise as to how the graph changes when you increase q and when you decrease q .

7. Ensure that a is NOT 0. Adjust q so that $q = 0$. Adjust the value of p so that $p > 0$.

a) Describe how the function text change as you increase p . (Make p bigger than 0).

b) Describe how the graph changes as you increase p .

c) Did you expect the change that you saw in 7b? Describe what is surprising?

8. Ensure that a is NOT 0. Adjust the value of p so that $p < 0$.

a) Describe how the function text change as you decrease p . (Make p less than 0).

b) Describe how the graph changes as you decrease p .

c) Did you expect the change that you saw in 8b? Did anything surprise you?

d) Look at your answer to question 3? Did the graph change when you adjusted p in question 3? Explain

9. Pick any values of a, p and q so long as a is not 0.

a) Look at the coordinates of the turning point of the graph.

b) Look at the values of p and q .

c) How does the turning point link to the values of p and q ?

10) Pick other values of a, p and q . Repeat 9a to 9c.

11) Write down 2 properties of a quadratic function that a changes. Describe how they change as a changes.

12) Write down 2 properties of a quadratic function that p changes. Describe how they change as p changes.

13) Write down 2 properties of a quadratic function that q changes. Describe how they change as q changes.

Homework – sketching a parabola of the form $y = a(x + p)^2 + q$

1. Pick any 3 values of a , p and q .
2. Without using Geogebra or a calculator plot the turning point of the function
3. Without using Geogebra or a calculator plot the y -intercept of the function
4. Without using Geogebra plot the x -intercepts of the function. You can use a calculator to help you solve the equation if necessary.
5. Draw the function
6. Check your answer using Geogebra.

Repeat steps 1-6 above so that you have drawn 4 of your own functions. Use different combinations of positive and negative a , p and q values.